

Feedforward Filters

General Form $y_t = x_t + a x_{t-1}$ $H(\omega) = 1 + a e^{-i\omega}$ $H(z) = 1 + a z^{-1}$ $ H(\omega) = \sqrt{1 + a^2 + 2a \cos(\omega)}$ $\frac{1}{2}$	Transfer Function $y_t = x_t + a x_{t-1}$ $\mathbf{Y} = \mathbf{X}(1 + a z^{-1})$ $\frac{\mathbf{Y}}{\mathbf{X}} = (1 + a z^{-1})$ $H(z) = 1 + a z^{-1}$...
Inverse Comb $y_t = x_t + R^L x_{t-L}$ $H(\omega) = 1 + R^L e^{-i\omega L}$ $H(z) = 1 + R^L z^{-L} = \frac{z^L - R^L}{z^L}$ $ H = 2 \sin(\omega L/2) $	Zeroes $z^L = R^L$ Poles $z = 0$ $z = R e^{i\theta}$ $e^{iL\theta} = 1$ $\frac{L\theta}{2\pi} = k$
Phase Response $[R_\omega e^{i\theta\omega}] \quad R_\omega$: MAGNITUDE RESPONSE $\theta_\omega = \tan^{-1}(\frac{\text{ImaginaryPart}}{\text{RealPart}})$ $\theta_\omega H(\omega) = e^{i\omega} (1 + a e^{-i\omega})$ $e^{i\theta\omega} = \frac{H(\omega)}{ H(\omega) }$	
$y_t = x_t - 2x_{t-1}$ $H(\omega) = 1 - 2e^{-i\omega} = \frac{e^{i\omega} - 2}{e^{i\omega}}$	$ H(\omega) = [5 - 2\cos(\omega)]^{1/2}$ $\theta_\omega = \tan^{-1}(\frac{2\sin(\omega)}{1 - 2\cos(\omega)})$
Palindromic Phase Response ODD TERMS $H(z) = 2 + z^{-1} - 3z^{-2} + z^{-3} + 2z^{-4}$ $H(\omega) = 2 + e^{-i\omega} - 3e^{-2i\omega} + e^{-3i\omega} + 2e^{-4i\omega}$ $H(\omega) = e^{-i2\omega} [4\cos(2\omega) + 2\cos(\omega) - 3]$ $ H(\omega) = 4\cos(2\omega) + 2\cos(\omega) - 3$ $\theta_\omega = -2\omega$	
Even Terms $y_t = x_t + \frac{1}{3}x_{t-1} + \frac{1}{3}x_{t-2} + x_{t-3}$ $H(\omega) = 1 + \frac{1}{3}e^{-i\omega} + \frac{1}{3}e^{-2i\omega} + e^{-3i\omega}$ $H(\omega) = e^{-3i\frac{\omega}{2}} [e^{3i\frac{\omega}{2}} + \frac{1}{3}e^{i\frac{\omega}{2}} + \frac{1}{3}e^{-i\frac{\omega}{2}} + e^{-3i\frac{\omega}{2}}]$	
Collect Cos with $e^{i\frac{\omega}{2}} + e^{-i\frac{\omega}{2}} = 2\cos(\frac{\omega}{2})$ Phasors	
$f(t) = e^{i\omega t}$ ω = angular freq (rad/s) Frequency: $f = \frac{\omega}{2\pi}$ Hz	Period: $T = \frac{2\pi}{\omega} = \frac{1}{f}$ sec Function $f(z) = \alpha z$ rotates by $\arg(\alpha)$, scales by $ \alpha $
Aliasing Sample rate: f_s Hz Nyquist frequency: $f_s/2$	$f_{alias} = f - n f_s $ (choose n to minimize)

Feedback Filters

General Form $y_t = x_t + R y_{t-1}$ $ H(\omega) = \frac{1}{ 1 - R e^{-i\omega} }$	$H(z) = \frac{1}{1 - R z^{-1}} = \frac{z}{z - R}$ Pole at $z = R$ (stable if $ R < 1$)
Reson Filter (2-Pole) $y_t = A_0 x_t + 2R \cos \theta \cdot y_{t-1} - R^2 y_{t-2}$ $H(z) = \frac{1}{(z - R e^{i\theta})(z - R e^{-i\theta})}$ $ H(\omega) \approx \frac{1}{\sqrt{1 - R^2}}$ at resonance Poles on unit circle at $z = R e^{\pm i\theta}$	
Gain factor $A_0 = (1 - R^2) \sin \theta$ Pole angle $\theta = \frac{2\pi \psi}{f_s}$ $\cos(\psi) = \frac{1 + R^2}{2R} \cos(\theta)$	... Pole Radius $R \approx 1 - B/2$ Bandwidth $B \approx 2(1 - R)$
Plucked String Filter $y_t + a y_{t-1} = a x_t + x_{t-1}$ $H(z) = \frac{a + z^{-1}}{1 + a z^{-1}}$ $ H(\omega) \rightarrow const$ (phase does not affect magnitude)	
Delay $L = \lfloor P \rfloor$ Period $P = \frac{f_s}{f_0}$ $\delta = P - L$ $\frac{f_s}{f} = L + \frac{1}{2} + \delta$	Low-pass $y = \frac{1}{2}(x_t + x_{t-1})$ $[\frac{1}{2} delay]$
All-Pass $a = \frac{1 - \delta}{1 + \delta}$ $\theta_\omega = 2\theta$ $\tan^{-1} x \approx x$ $\theta_\omega = 2(\frac{a-1}{a+1}\omega)$	
Phase Delay $\delta = \frac{2}{\omega} \tan^{-1}[\frac{a-1}{a+1} \tan(\omega/2)]$ $\tau(\omega) = \frac{-2 \arctan(\frac{a \sin \omega}{1 + a \cos \omega})}{\omega}$	
Comb Filter [HARMONIC SERIES] $H(z) = \frac{1}{1 - R^L z^{-L}}$ $ H = 2 \cos(\omega L/2) $	
Peaks $\omega = 2\pi k/L$ Zeroes $1 - R^L z^{-L} = 0$	Poles $-1 = e^{i\pi}$ $z^L = -1$
Impulse Response Sum $\sum_{t=0}^{\infty} h_t = \frac{1}{1 - R^L}$	

Periodic Sound

TIME DOMAIN SIGNALS CONSIDERED TO BE PERIODIC FUNCTIONS $f(t)$ TIME DOMAIN $\neq$ FREQUENCY DOMAIN	
$\omega_0 = \frac{2\pi}{T}$ $f(t) = e^{ik\omega_0 t}$ $\int e^{at} = \frac{1}{a} e^{at} + C$	$\int e^{iat} = \frac{1}{ia} e^{iat}$ $f(t + kT) = f(t)$ $\frac{1}{i} = -i$ $\frac{1}{-i} = i$
Inner Product $\frac{1}{T} \int_0^T f(t) g(t) dt$ $\langle u, v \rangle = \sum_n \bar{u}_n v_n$...	
Forward Fourier $C_k = \frac{1}{T} \int_0^T f(t) e^{ik\omega_0 t} dt$...	
Inverse Fourier $f(t) = C_0 + 2 \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) - b_k \sin(k\omega_0 t)]$	
DFT & Fourier Basis Basis Vectors: $u_k[n] = e^{i2\pi k n/N}$ Orthogonality: $u_k \cdot u_j = N \delta_{kj}$	
$X[k] = \sum_{n=0}^{N-1} x[n] e^{-i2\pi k n/N}$	
$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{i2\pi k n/N}$	
Coefficient: $c_k = X[k]/N$	
Square Wave Fourier Period T , 50% duty cycle:	
$c_k = \begin{cases} 0 & k \text{ even} \\ \frac{4}{k\pi} & k \text{ odd} \end{cases}$	
General Fourier: $c_k = \frac{1}{T} \int_0^T s(t) e^{-i2\pi k t/T} dt$	
Calculations Quantization ROOT-MEAN-SQUARE, SIGNAL-TO-NOISE RATIO	
RMS Error $[a, b] \rightarrow \frac{1}{b-a} \int_a^b f(x) dx$ $[0, \frac{1}{2}] \rightarrow \frac{1}{0.5} \int_0^{0.5} x^2 dx = \frac{1}{\sqrt{12}}$	SNR $-2^{B-1} \leq x_t \leq 2^{B-1} - 1$ $2(2^{B-1}) = 2^B$ $\frac{2^{B-1}}{RMS} = 2^B \sqrt{3}$
$[-0.5, 0.5] \rightarrow \frac{1}{2\sqrt{3}} \approx 0.289$	
dB = $20 \log_{10}(A)$	

Unit Circle $e^{i\pi/2} = i, e^{i\pi} = -1,$ $e^{i3\pi/2} = -i, e^{i2\pi} = 1$ $e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i,$ $e^{i2\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ $e^{i\pi/4} = \frac{1+i}{\sqrt{2}},$ $e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$	Complex Arithmetic $z = a + bi = r e^{i\theta},$ $r = z = \sqrt{a^2 + b^2}$ $i^2 = -1, i^3 = -i$ $i^4 = 1, i^5 = i$								
Operations $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$ $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$ $(re^{i\theta})^n = r^n e^{in\theta}$ $(a + bi)(a - bi) = a^2 + b^2$ Conjugate: $\bar{z} = a - bi = r e^{-i\theta}$ Rotation by θ : multiply by $e^{i\theta}$ Trig Identities	... $\cos(2a) = \cos^2(a) - \sin^2(a)$ $\sin(2a) = 2\sin(a)\cos(a)$								
$ 1 + e^{i\omega} = 2 \cos(\omega/2) $ $ 1 - e^{i\omega} = 2 \sin(\omega/2) $ $e^{it} - e^{-it} = 2i \sin(t)$ $e^{it} + e^{-it} = 2 \cos(t)$	Quadratic from Complex Roots $(x - (a + bi))(x - (a - bi)) = x^2 - 2ax + (a^2 + b^2)$ Cubic Factoring $x^3 + 1 = (x + 1)(x^2 - x + 1)$ $x^3 - 1 = (x - 1)(x^2 + x + 1)$								
Taylor Series $e^{ix} = 1 + i\frac{\pi}{2} + \frac{i(\frac{\pi}{2})^3}{3!} + \dots$	Product $\prod_{k=1}^n e^{ik\theta} = e^{i\theta n(n+1)/2}$ Geometric Series $\sum_{k=0}^n e^{ik\theta} = \frac{1 - (e^{i\theta})^{n+1}}{1 - e^{i\theta}}$								
	+3dB = $\sqrt{2}$ × amplitude +6dB = 2× amplitude +20dB = 10× amplitude								
Filter Combinations • Cascade: $H_{total} = H_1 \cdot H_2$ • Parallel: $H_{total} = H_1 + H_2$ • Feedback: $H_{total} = \frac{GH}{1 - FH}$									
Phase Properties Linear Phase: • Symmetric coefficients • $\phi(\omega) = -\omega D$ where D = group delay • No distortion of wave shape Phase Delay vs Group Delay: • Phase delay: $\tau_p = -\phi(\omega)/\omega$ • Group delay: $\tau_g = -d\phi/d\omega$									
<table><tr><th>Operation</th><th>Formula</th></tr><tr><td>Complex dot</td><td>$(a_1, b_1) \cdot (a_2, b_2) = \bar{a}_1 a_2 + \bar{b}_1 b_2$</td></tr><tr><td>Parseval</td><td>$\frac{1}{T} \int s ^2 dt = \sum c_k ^2$</td></tr><tr><td>Impulse</td><td>$\delta[n] = \{1 \text{ if } n = 0, 0 \text{ else}\}$</td></tr></table>	Operation	Formula	Complex dot	$(a_1, b_1) \cdot (a_2, b_2) = \bar{a}_1 a_2 + \bar{b}_1 b_2$	Parseval	$\frac{1}{T} \int s ^2 dt = \sum c_k ^2$	Impulse	$\delta[n] = \{1 \text{ if } n = 0, 0 \text{ else}\}$	
Operation	Formula								
Complex dot	$(a_1, b_1) \cdot (a_2, b_2) = \bar{a}_1 a_2 + \bar{b}_1 b_2$								
Parseval	$\frac{1}{T} \int s ^2 dt = \sum c_k ^2$								
Impulse	$\delta[n] = \{1 \text{ if } n = 0, 0 \text{ else}\}$								